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Data Science



FramePerfect - Video Editing Enhancer

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Project Guide
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Outline

- Introduction
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Introduction

This application aims to improve the video editing experiences using AI and providing automation for features such as subtitles generation, assets recommendation and work video segmentation and extraction of information (like sentiment analysis) from video type of data.

Motivation:

- **Manual Video Editing is Time-Consuming** – Editors spend hours on tasks like subtitles and segmentation. AI automation speeds up workflows and reduces effort.
- **Rising Demand for AI-Powered Editing** – Digital creators need smarter tools for effortless video enhancement. AI bridges the gap between professional editing and accessibility.
- **Supporting Small Content Creators** – Many lack editing skills or budgets for professionals. This tool helps them create high-quality videos with ease.

Objectives

- 1) To provide easier extraction and integration of subtitles using AI based speech-to-text models (Whisper)
- 2) To Extract Insights from Transcript content to detect mood using NLP.
- 3) To Perform Sentiment Analysis on Audio Content using NLTK approach.
- 4) To enhance video editing workflow by giving contextual assets using extracted insights using NLP

Literature Survey of the existing system

SR NO	Title	Author	Year	Methodology	Result
[1]	Subtitle Synchronization Using Whisper ASR Model	T. P, M. Azneed, M. Sanas, J. P. P. M, H. P. Naik and D. B	2014	The process uses FFmpeg for audio extraction, Whisper ASR for speech recognition, and text comparison for precise subtitle synchronization by adjusting timestamps between input and transcribed SRT files.	The system successfully enhances subtitle synchronization accuracy, effectively adjusting timestamps even with varying audio delays.

SR NO	Title	Author	Year	Methodology	Result
[2]	Sentiment Analysis of Speeches Using VADER, Sentiment Intensity Analyzer, and TextBlob.	Ankush Joshi, Deepika Joshi, Vikash Kumar, and Anuj Kumar	2024	It employ a hybrid NLP-based sentiment analysis approach by transcribing speeches using ASR models and analyzing sentiment using VADER, TextBlob, and Sentiment Intensity Analyzer. The study extracts emotion-related features and classifies sentiments using machine learning models.	VADER excels in short texts, TextBlob captures contextual sentiment, and Sentiment Intensity Analyzer quantifies emotional intensity, making the system effective for speech sentiment profiling.

SR NO	Title	Author	Year	Methodology	Result
[3]	Conversion of Speech to Text: Transcribe and Translate Using Whisper AI	Dr. Shilpa V1, Chandan Kumar, Girija V3 Assistant Professor, Dept of CSE, Cambridge Institute of Technology	August 2024	The methodology involves data preprocessing, model training, and deployment for AI-powered video editing. A large-scale dataset of 680,000 hours of audio is curated, filtered, and segmented for training Whisper's encoder-decoder transformer model	Whisper AI was evaluated against standard speech recognition benchmarks, achieving competitive results without task-specific fine tuning. The model's performance was compared to: : Deep Speech Wav2Vec 2.0

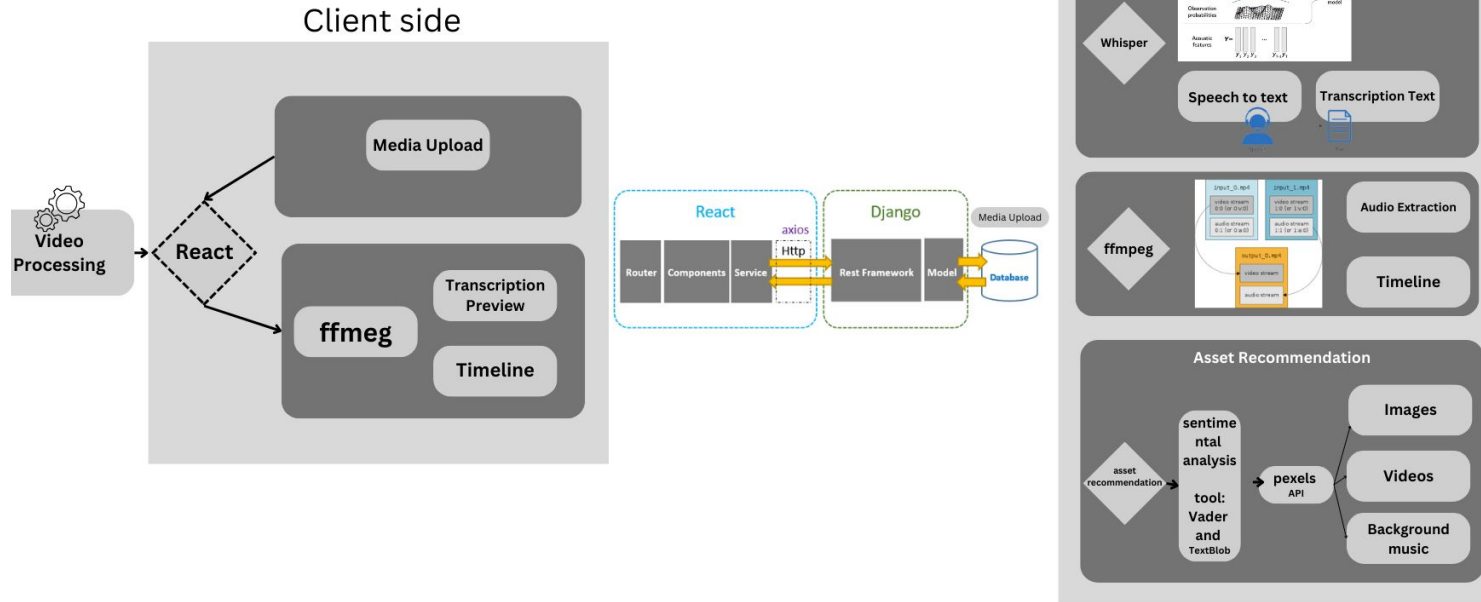
Limitations of existing systems

- **High Learning Curve & Cost** – Professional editing software requires extensive expertise and costly subscriptions, making it inaccessible for small creators who lack editing skills.
- **Lack of Accurate AI-Driven Subtitles** – Software offers AI-based transcription but struggles with multiple speakers, accents, and background noise, requiring manual corrections.
- **No Integrated Sentiment-Based Editing Insights** – AI-assisted editing software lacks deep sentiment analysis that can recommend background music, stock footage, or effects based on the mood of the content.

Problem Statement

Video editing is time-consuming and requires manual effort for tasks like subtitles, segmentation, and asset selection. Many small creators lack the skills or budget for professional editing. This project aims to develop an AI-powered tool that automates these processes using Whisper for speech-to-text, NLP for sentiment analysis, and CV for scene detection, making editing efficient and accessible.

System Design



Technologies and methodologies

Frontend (React.js)

- React Player for video playback.
- Wavesurfer.js for waveform timeline & subtitle alignment.
- Editable Subtitle Component to allow real-time text corrections.
- FFmpeg.wasm for in-browser video processing & export.

Backend (Django)

- Django REST API for handling video uploads & processing requests.
- Whisper Model (OpenAI) for high-accuracy speech-to-text conversion.
- FFmpeg (server-side) for advanced video processing & embedding subtitles.

Database

- Uploaded video metadata (title, duration, format).
- Subtitle data (timestamps, edited subtitles, confidence scores).
- User editing history for undo/redo functionality.

Implementation



Menu

+

Cut

Media Preview

Video #1

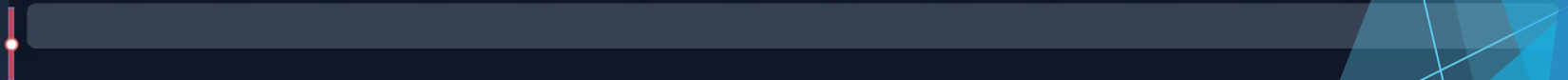


Transcription

Yes, after the ML module detects a suspicious website, it triggers this automation and scheduling batch made in Apache Airflow with Clumps bunch of Python scripts together. It will essentially automate the process of reporting to the authorities as well as logging it to the SAS platform and generating status report for stakeholders. This whole architecture will be continuously used by Docker with separated API endpoints and it makes it scalable and easy for

Timeline

0s 2s 4s 6s 8s 10s 12s 14s 16s 18s 20s 22s 24s 26s 28s 30s



Novelty : Rule Based Real Time Sentiment Algorithm

- RBRTS is a simple implementable Keyword extraction and Asset Recommendation Algorithm
- Traditional Recommendation Algorithms are slow and require bigger context, which makes them unsuitable for Real time Streaming

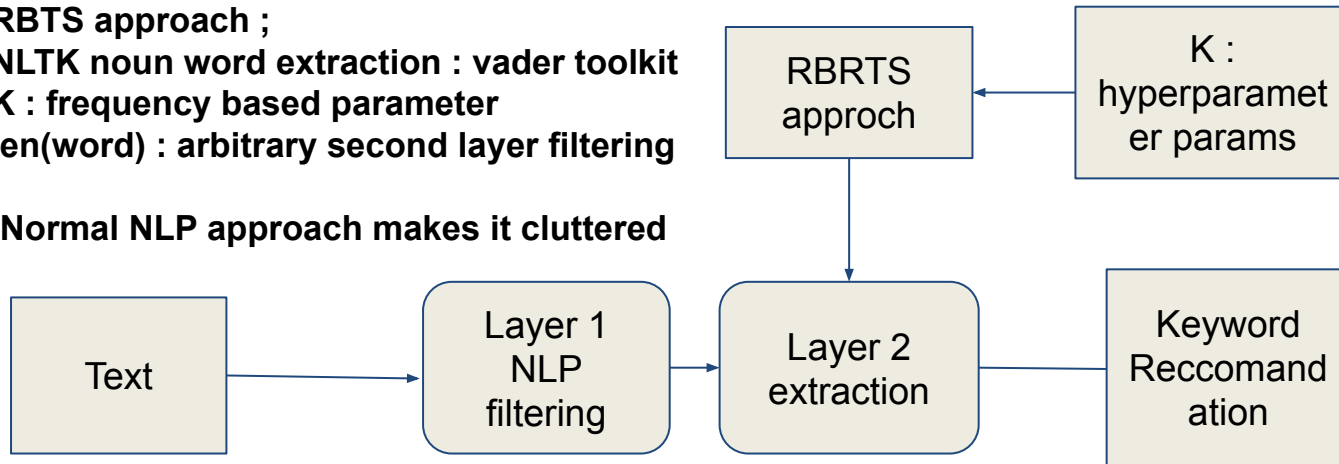
RBTS approach ;

NLTK noun word extraction : vader toolkit

K : frequency based parameter

len(word) : arbitrary second layer filtering

Normal NLP approach makes it cluttered



Conclusion

Automated AI-driven video editing reduces manual effort in subtitle generation, segmentation, and asset recommendations. By integrating NLP, computer vision, and Whisper AI, it enhances video processing accuracy and efficiency. This technology empowers small content creators and YouTubers by providing professional-grade editing without requiring advanced skills. It makes video editing more accessible, intelligent, and time-efficient, bridging the gap between manual and AI-assisted workflows.

References

- [1] T. P. M. Azneed, M. Sanas, J. P. P. M, H. P. Naik and D. B, "Subtitle Synchronization Using Whisper ASR Model," 2024 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS), Chennai, India, 2024, pp. 1-6, doi: 10.1109/ICPECTS62210.2024.10780268.
- [2] Joshi, A., Joshi, D., Kumar, V., & Kumar, A. (2024). Sentiment Analysis of Speeches Using VADER, Sentiment Intensity Analyzer, and TextBlob. Proceedings of the 4th International Conference on Advancement in Electronics & Communication Engineering (AECE), November 2024. DOI:
- [3] S. Wang, C. -H. Yang, J. Wu and C. Zhang, "Can Whisper Perform Speech-Based In-Context Learning?," ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Seoul, Korea, Republic of, 2024, pp. 13421-13425, doi: 10.1109/ICASSP48485.2024.10446502.

**Thank
You...!!**